

CYPHER

Technical Whitepaper — Version 1.4

Quantum-Resistant Privacy Blockchain with Cytherpunk Inclusion Proof

Token: \$CYPHER · Total Supply: 1,000,000,000,000 \$CYPHER (research-stage target)

Consensus: CytherPOW (Equihash-512) · Emission: research-stage, ~5-year schedule

Core Primitive: CIP — Cytherpunk Inclusion Proof (FRI-based STARK, Poseidon2)

Quantum Shield: CRYSTALS-Dilithium5 (NIST FIPS 204)

“Ghost chains don’t Exist.”

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§1 — Abstract

Cypher is a quantum-resistant, privacy-first Layer-1 blockchain, currently at the research and design stage. Its core claim, Proof of Existence (PoE), treats every transaction as a verifiable existence event. In this revision, the cryptographic core has been consolidated: rather than combining a lattice-adapted ring signature with a separate Groth16 zk-SNARK, Cypher now proves existence and privacy together inside a single FRI-based STARK circuit — the **Cyberpunk Inclusion Proof (CIP)**.

CIP proves four statements simultaneously: miner-set Merkle membership, nullifier freshness, balance validity, and block-height existence binding. A CRYSTALS-Dilithium5 signature (NIST FIPS 204) wraps the proof as an outer envelope. Because STARKs rely on collision-resistant hashing (Poseidon2) rather than elliptic-curve or trusted-setup assumptions, CIP removes the post-quantum caveat that applied to the BLS12-381-based privacy layer in the prior design, and removes the need for a Powers-of-Tau ceremony.

1,000,000,000,000 \$CYPHER is planned for emission to miners over a five-year, research-stage schedule via CypherPOW (Equihash-512). A foundation mining allocation of 40–60% of early supply is disclosed directly in this document rather than described as a fair launch — see §12.

The native ecosystem — Mask Wallet, MaskSwap CLMM DEX, Cypher Bridge, CypherOracle, ShieldLend, and Cypher DAO — is designed to operate on CIP-attested, privacy-preserving state transitions, with a Zcash-style user-controlled viewing key model for selective disclosure. This document is a living research artifact; it carries no delivery dates and should be read alongside §18, Limitations & Future Work.

§2 — Introduction & Motivation

The blockchain industry faces two converging threats. **Quantum vulnerability:** ECDSA and RSA-2048 are expected to be broken by sufficiently large quantum computers via Shor's algorithm; Bitcoin, Ethereum, Monero, and Zcash all rely on elliptic-curve cryptography. NIST finalised its first post-quantum standards in August 2024 — the threat timeline is real, if not immediate. **Privacy collapse:** on-chain analytics can trace the majority of transactions on transparent chains, and even Monero's ring signatures and Zcash's zk-SNARKs carry known deanonymisation vectors or rest on quantum-vulnerable elliptic-curve foundations.

Cypher's design goal is to survive both threats at once: CRYSTALS-Dilithium post-quantum signatures (NIST FIPS 204) for all outer-envelope signing; a unified FRI-based STARK circuit (CIP) for existence-and-privacy proofs, in place of separate elliptic-curve-based primitives; and GPU-mineable proof-of-work for decentralised block production.

§3 — Proof of Existence — Core Design

PoE remains Cypher's foundational claim: every transaction is validated as an existence event tied to a specific block height. In this revision, the mechanism that produces that proof has moved from a standalone ring signature to the CIP circuit described in §4, which proves existence membership as one of its four statements.

3.1 — Layered Architecture (revised)

Layer	Component	Technology	Role
L2 Applications	MaskSwap · ShieldLend · Oracle · DAO	DeFi	All CIP-attested
L1 ★ CIP	Cytherpunk Inclusion Proof	FRI-STARK (Poseidon2) + Dilithium5 envelope	Quantum-safe existence + privacy attestation
L0 Consensus	CytherPOW	Equihash-512	Block production · GPU hashrate

The prior four-layer model (L3/L2/L1/L0) separated privacy (Groth16) from existence attestation (ring signatures) into two proof systems. Folding both into one STARK circuit removes the need to reconcile miner-side attestation against user-side privacy proofs after the fact — they are now one statement, checked once.

§4 — Cytherpunk Inclusion Proof (CIP)

CIP is Cyther's unified existence-and-privacy primitive: a single FRI-based STARK circuit proving four statements about a transaction simultaneously.

4.1 — The Four Statements

- **Miner-set Merkle membership** — the transaction is attested by a member of the active miner set at the relevant block height (replaces the prior ring-signature construction).
- **Nullifier freshness** — the spent output's nullifier has not appeared before (double-spend prevention).
- **Balance validity** — inputs equal outputs under Pedersen-style commitments, with no negative amounts.
- **Block-height existence binding** — the proof is cryptographically bound to a specific block height and chain state, so it cannot be replayed at a different point in the chain.

4.2 — Structure

```
CIP_proof = STARK.Prove(circuit_CIP, public_inputs, private_witness)

public_inputs = (miner_root, nullifier, height, chain_state)
private_witness = (sk, amount, randomness, merkle_path, miner_index)

circuit_CIP asserts:
[1] MerkleVerify(pk_signer, miner_root, merkle_path) == true
[2] nullifier == Poseidon2(sk) and nullifier not in NullifierSet
[3] Pedersen(amount_in) == Pedersen(amount_out) + fee
[4] event_id == Poseidon2(payload, height, chain_state)

envelope = Dilithium5.Sign(sk_infra, CIP_proof || event_id)
```

4.3 — Why a Single Circuit

The prior design signed existence with a lattice-adapted ring signature and proved privacy separately with a Groth16 zk-SNARK. Reconciling the two required the miner-side attestation (who produced this block?) and the user-side privacy proof (what does this transaction actually move?) to agree on the same transaction without either proof system ever checking the other's statement directly. CIP resolves this by proving both inside one circuit, over one set of public inputs, verified by one algorithm.

4.4 — Properties Summary

Property	Specification
Proof system	FRI-based STARK
Hash function	Poseidon2
Trusted setup	None required (transparent setup)
Anonymity model	Logarithmic scaling across all chain participants, not a sampled ring
Statements proven	4 (membership, freshness, balance, existence binding)
Envelope signature	CRYSTALS-Dilithium5 (NIST FIPS 204)
Quantum safety	Hash-based; no known Shor-level break, only Grover-bounded speedup

4.5 — Improvement Over the Prior Ring Design

Anonymity-set scaling improves from a sampled ring of $n = \max(11, \lceil \log_2(\text{miners}) \rceil \times 3)$ — 11 to 60 members depending on network size — to a logarithmic set across *all* chain participants. This is a genuine anonymity improvement, not a rebrand of the same mechanism.

4.6 — Caveats

CIP is a research proposal without audited precedent. Poseidon2 is a newer hash function with a shorter cryptanalysis track record than SHA-256 or Keccak. STARK proof sizes for a four-statement circuit at Cypher's parameters have not yet been benchmarked on real hardware — see §18.

4.7 — Concrete Parameters (proposed)

Prior revisions left the field, hash, and FRI parameters unstated — without them, nobody can estimate CIP's actual proof size, prover time, or soundness level. The following is a starting proposal, modelled on parameter choices published by comparable production STARK provers, and is explicitly not yet audited or benchmarked.

Parameter	Proposed Value	Rationale
Base field	Goldilocks, $p = 2^{64} - 2^{32} + 1$	64-bit, FFT-friendly; used in production STARK provers
FRI extension degree	Quadratic extension of base field	Needed to reach adequate soundness from a 64-bit base field

Parameter	Proposed Value	Rationale
Blowup factor (rate)	1/8	Standard production range; trades proof size against prover time
FRI query count	84	Targets conjectured soundness in the 90–100-bit range with grinding
Grinding (PoW) bits	20	Boosts soundness beyond the pure query-count bound
Hash (in-circuit & Merkle)	Poseidon2, same field	Arithmetisation-friendly; consistent with §4.6 caveat

These are proposed starting parameters, not a finalised specification. They still require independent cryptanalysis and full-circuit benchmarking before CIP can be considered production-ready — §4.7.1 below is a first, narrow data point, not that benchmarking.

4.7.1 — A Real Benchmark at These Parameters

Rather than leave every number above as an unverified guess, we ran a real STARK prover (Winterfell) at exactly these parameters — the Goldilocks field, blowup 8, 84 queries, grinding 20 — using Rescue-Prime as a stand-in for Poseidon2 (both are algebraic, arithmetisation-friendly hashes from the same design lineage; Poseidon2 itself isn't available in this library). This is a toy 2-column computation, *not* the real CIP circuit, which needs Merkle verification, nullifier and balance checks, and a hash evaluation in far more columns. It answers one narrow question: do proofs at this field, hash family, and query count actually run, and what do they look like when they do.

Trace rows	Prove time (1 core)	Verify time	Proof size
512	4.4 s	11.2 ms	43.6 KB
2,048	4.9 s	16.8 ms	72.0 KB
8,192	7.6 s	23.4 ms	101.4 KB

Single-threaded, one CPU core, release build, all proofs verified correctly. Real hardware would very likely run multi-core; this is a pessimistic, not optimistic, number.

Two things stand out. First, the proof sizes (44–101 KB) land right in the “tens of kilobytes” range §4.8 already assumed — that assumption holds up. Second, prove time is seconds, not milliseconds, for a circuit with only 2 columns. The real CIP circuit will have far more, and recursion (§4.8) needs a block producer to compose many of these into one proof inside a 120-second block window. That timing question, not proof size, looks like the real risk to throughput — see §7.1 and §18.

4.8 — Recursion & Light Clients

A standalone STARK proof is, like other STARK proofs, larger than the old ring signature — typically tens of kilobytes rather than the ~4.6 KB the prior ring design used, confirmed directly by the benchmark in §4.7.1. Left unaddressed, that would make blocks larger and full verification slower as the chain grows. Cypher proposes closing that gap with **recursive proof composition**: a block producer recursively verifies every per-transaction CIP proof inside a single block-level STARK proof, and each new block's proof recursively verifies the previous block's proof, forming a single chain-of-proofs. This is the same general approach used in production by Mina's Pickles (which keeps Mina's entire chain state verifiable at roughly 22 KB) and by recursive STARK provers such as Plonky2-class systems.

With recursion, a light client — a mobile wallet or a browser — only needs to verify one constant-size (or slowly growing) recursive proof to be convinced of the chain's full validity, rather than downloading and replaying transaction history. Individual per-transaction proofs are still propagated at their pre-recursion size during mempool gossip, since aggregation happens at block-production time; that mempool bandwidth cost is real and still needs benchmarking. More pressing, per §4.7.1: recursion itself takes real prover time per transaction composed, and whether a block producer can compose enough transactions inside 120 seconds to hit any meaningful throughput is now the open question that matters most — see §18.

§5 — Quantum Resistance: CRYSTALS-Dilithium (NIST FIPS 204)

All Cypher signing operations use CRYSTALS-Dilithium, standardised as NIST FIPS 204 (August 2024) and provably secure under MLWE and MSIS hardness assumptions, resistant to both classical and quantum adversaries.

Parameter Set	NIST Level	PQ Security	Pub Key	Sig Size	Usage in Cypher
Dilithium2	Level 2	128-bit PQ	1,312 B	2,420 B	Wallet-layer user signing
Dilithium3	Level 3	192-bit PQ	1,952 B	3,293 B	Validators & oracle relayers
Dilithium5	Level 5	256-bit PQ	2,592 B	4,595 B	CIP outer envelope signature ★

Dilithium5 wraps every CIP proof at the infrastructure layer — secure against adversaries with large-scale fault-tolerant quantum computers, not only near-term NISQ devices.

§6 — PoE Integration & Transaction Model

6.1 — Transaction Flow

```

Step 1: User constructs transaction locally (amount, recipient, randomness).
Step 2: User/wallet generates CIP_proof = STARK.Prove(...), bound to the
        current miner_root and chain_state.
Step 3: User signs: user_sig = Dilithium2.Sign(sk_user, CIP_proof || tx_meta)
Step 4: Submit { encrypted_payload, CIP_proof, user_sig } to mempool.

```

Block producer (per block):

```

Step 5: poe_root = MerkleRoot({ event_id_i || CIP_proof_i } for all i)
Step 6: block.hash = Equihash512.Solve(header || poe_root || nonce)

```

Note: a block missing a valid CIP proof for any transaction is INVALID.
CIP verification is consensus-enforced.

6.2 — Event Finality

Events are final at 6-block confirmation depth. Tampering with any event invalidates the block's PoW, since poe_root is committed in the header. Reverting 6+ blocks requires >51% hashrate *and* valid CIP proof forgery simultaneously — an infeasible compound attack under any known classical or quantum adversary model.

This is a *cost* claim, not a structural impossibility — worth stating plainly after a logic-level simulation of the chain-selection rule (§19.2) confirmed that a sufficiently long, heavier chain can still out-compete and reorg past 6 confirmations by construction. What makes 6 blocks practically final is that building such a chain requires sustained majority hashrate, not that the rule forbids it.

§7 — CypherPOW Consensus (Equihash-512)

CypherPOW uses Equihash($n=512$, $k=9$), a memory-hard proof-of-work requiring roughly 2.7 GB RAM per solver instance. Memory hardness resists ASIC dominance and preserves GPU miner decentralisation. Block target: 120 seconds.

```

Block Header: version || prev_hash || poe_root || merkle_root || timestamp
              || nBits || nonce

```

Validity Rules (all must pass):

```

[1] Equihash512.Verify(header, solution) == true
[2] H(header) < target(nBits)
[3] poe_root == MerkleRoot(all CIP proofs in block)    // consensus rule
[4] Every CIP proof passes circuit verification
[5] Coinbase signed with miner Dilithium2 key

```

Difficulty Adjustment (every 500 blocks):

```

D_new = D_old x (T_target x N) / sum(t_i),  N = 500, T_target = 120s

```

7.1 — Throughput & Comparative Benchmarks

Prior revisions gave a single “10–60 tx/s” range that, checked directly, didn’t actually fall out of any one consistent scenario. Recomputing it as two separate, labelled bounds instead of one blended guess:

Scenario	Per-tx size	At ~4MB block body	Implied throughput
Recursion-compressed (on-chain footprint only)	~300–600 B	~7,000–14,000 tx/block	~58–117 tx/s
Pre-recursion (individual CIP proof)	~40–150 KB	~30–100 tx/block	~0.2–0.85 tx/s

The first row is the number that matters once recursion (§4.8) is actually working; the second is what happens if a block only carries un-aggregated individual proofs. Neither is a promise — both are order-of-magnitude, byte-size-only bounds.

That said, byte size is very likely not the real bottleneck. The benchmark in §4.7 measured 4.4–7.6 seconds of single-core prove time for a trivial 2-column toy circuit — not the real CIP circuit, which needs Merkle verification, nullifier and balance checks, and a hash evaluation, and will have far more columns. Recursion (§4.8) requires a block producer to compose every included transaction’s proof into one block-level proof inside the 120-second block window. If per-transaction proving genuinely costs seconds rather than milliseconds, *prover throughput* — not block body size — sets the real ceiling, and it could be well below the byte-size figures above. This needs the real circuit benchmarked, and multi-core / hardware-accelerated proving measured, before either row in the table means anything operationally — see §18.

Network	Block Time	Typical / Sustained Throughput	Privacy
Monero	2 min	~0.3–0.5 tx/s sustained (~20–40K tx/day)	Mandatory (ring signatures)
Zcash (Orchard)	~75s	Lower than transparent pool; shielded-pool adoption still growing	Opt-in shielded pool
Solana	~0.4s slot	On the order of 1,000–4,000 tx/s sustained in practice	None
Cypher (CIP, proposed)	120s	58–117 tx/s if byte-bound; far lower if prover-bound (see above)	Mandatory (CIP)

Monero figures derived from public network statistics; Solana figures are widely-cited practical ranges, not theoretical maximums. Cypher’s 120s block time is slow relative to non-private, non-PoW chains; see the GHOSTDAG note below and §18.

Open design thread: Kaspas’s GHOSTDAG is being studied as a reference for faster, parallel block production that could improve on the 120s figure above while keeping Equihash’s ASIC-resistance property; no change to CypherPOW has been adopted yet — see §18.

§8 — Privacy Model & Key Management

Cypher’s privacy model is mandatory by default: there is no public, plaintext chain state to opt out of. Every CIP-attested transaction hides amount, sender, and receiver by construction (§4).

8.1 — Viewing Keys

Selective disclosure follows a Zcash-style, user-controlled viewing key model: a user may derive and share a viewing key that lets a chosen third party (an auditor, an exchange, a tax authority) decrypt *that user's own* transaction history, without exposing any other participant's data and without granting on-chain visibility to anyone not holding the key.

8.2 — KYC Boundary

Know-your-customer checks are handled at fiat on/off-ramps — exchanges and bridges into fiat rails — rather than at the protocol layer. The chain itself performs no identity verification and stores no identity data.

8.3 — Rejected Design: Foundation-Held Universal Key

An earlier design considered a Foundation-held universal decryption key, letting the foundation decrypt any transaction for compliance purposes. This was declined and removed: a single universal key of this kind is a backdoor regardless of who holds it, and undermines the “reveals nothing about its users by design” claim made elsewhere in this document.

§9 — MaskSwap CLMM Protocol

MaskSwap is Cypher's native concentrated-liquidity AMM. Every trade, LP position, fee accrual, and range order is a CIP-attested event with full built-in privacy. Structural MEV prevention is provided via an encrypted mempool and batch-auction settlement.

CLMM (within active range $[p_a, p_b]$):

$dx = L \times (1/\sqrt{p}) - 1/\sqrt{p_b})$

$dy = L \times (\sqrt{p} - \sqrt{p_a})$

Fees: LP fee 0.25% -> range-active liquidity providers

Protocol fee 0.05% -> Cypher DAO treasury

Anti-MEV: Encrypted mempool * 12-second batch auction * zk-committed LP shares

Open design thread: LP yield mechanics for wrapped bridge assets (cETH, cBTC, cSOL) are not yet finalised — see §18.

§10 — Cross-Chain Bridge Architecture

The Cypher Bridge uses threshold ECDSA (tECDSA) with 8-of-15 MPC for asset custody on external chains. Validators sign internally with Dilithium5 and present ECDSA signatures externally. All bridge events are CIP-attested on the Cypher side. Supported chains: ETH, BNB, SOL, BTC.

Security: requires compromise of 8 of 15 independent validators simultaneously; internal communications are Dilithium3-signed.

10.1 — Upgrade Path: zk Light-Client Bridging

Federated MPC custody of this kind is the model behind several of the largest bridge losses in the industry to date (Ronin, Wormhole, Poly Network), because security reduces to trusting a fixed committee rather than the source chain's own consensus. Since Cypher already carries FRI-STARK proving infrastructure for CIP, the proposed upgrade path replaces federated trust with a **zk light-client bridge**: a STARK circuit that verifies the source chain's own consensus proof directly (for example, an Ethereum sync-committee signature circuit, or a Bitcoin SPV header-chain circuit), so that custody correctness reduces to “the source chain's consensus was not broken” rather than “8 of 15 validators are honest.”

This is a materially larger engineering effort than the current design — each supported chain needs its own light-client circuit — so it is proposed as a phased rollout starting with the highest-value corridor, with the current 8-of-15 tECDSA model remaining in place for chains not yet migrated. Lattice-based threshold MPC remains a separate, complementary upgrade for whichever corridors still rely on tECDSA, since tECDSA itself is classically secure but not post-quantum. Neither upgrade has a committed date — see §18.

§11 — CypherOracle

Cypher's native decentralised price oracle uses Winsorised weighted-median aggregation. All updates are CIP-attested events signed by Dilithium3 relayers.

```
Source 1 (50%): MaskSwap TWAP, 60-minute rolling window
Source 2 (40%): 15 independent Dilithium3-signed relayers, outliers Winsorised at 2-sigma
Source 3 (10%): Schelling-point commit-reveal (20+ assets)
Final:         WeightedMedian(TWAP, relayers, schelling)
Heartbeat:     60s normal, 5s on deviation > 0.5%
DeFi delay:    15-minute settlement for liquidations
```

11.1 — Relayer Decentralisation

Fifteen relayers is a small decentralised set relative to established oracle networks (Chainlink and Pyth both run considerably larger, stake-weighted node sets). Two changes are proposed: first, grow Source 2's relayer set well beyond 15, with stake-weighted selection and slashing for provably incorrect reports; second, add a fourth source that reads external DEX pool state directly through a zk light-client proof (reusing the bridge circuits from §10.1) rather than through a trusted relayer, reducing reliance on any single source type. Neither change is scheduled — see §18.

§12 — Token Economics & Emission Schedule

1,000,000,000,000 \$CYPHER is the hard-capped, planned total supply for a five-year, research-stage emission schedule via CypherPOW (Equihash-512).

12.1 — Foundation Allocation (disclosed)

A foundation mining allocation of 40–60% of early-period supply is part of this design. This document discloses that allocation directly, rather than describing the launch as a zero-premine, 100%-to-miners fair launch.

12.2 — Emission Formula

```

R_y = R_0 × k^y
R_0 = 1,522,069 $CYPHER    (Year 1 per-block reward)
k   = 0.625                (geometric decay factor)
y   = 0, 1, 2, 3, 4        (year index; 0 = first emission year)

```

12.3 — Corrected Schedule (v1.4)

Earlier revisions of this table stated round figures (400/250/175/100/75B, summing exactly to 1,000B) alongside the $R_y = R_0 \times k^y$ formula, and flagged an unresolved “~80% tail” without actually running the formula. Recomputing $R_y = R_0 \times k^y$ directly shows the table’s own year-1 and year-2 figures match the formula, but years 3–5 did not — they were round numbers picked to sum cleanly, not computed. The table below is the formula’s actual output.

Year	Block Reward (\$CYPHER)	Blocks	\$CYPHER Issued	Cumulative
Yr 1	1,522,069	262,800	400,000,000,000	40.00%
Yr 2	951,293	262,800	249,999,833,250	65.00%
Yr 3	594,558	262,800	156,249,895,781	80.62%
Yr 4	371,599	262,800	97,656,184,863	90.39%
Yr 5	232,249	262,800	61,035,115,540	96.49%

Total mined under this formula: 964,940,762,634 \$CYPHER — 96.49% of the 1,000,000,000,000 nominal figure. That figure is best read as a hard ceiling the schedule approaches, not a guaranteed issuance target: the formula, not a rounded table, is the source of truth for what actually gets mined.

This also corrects the previous estimate in this document, which guessed the tail converged to roughly 80% without computing it; the real figure is 96.49%. Both numbers were wrong until actually run — which is the point of §19.2 below.

12.4 — Supply Properties

Property	Value
Total planned supply	1,000,000,000,000 \$CYPHER — nominal hard cap; ~964.9B actually mined under §12.3’s schedule
Foundation mining allocation	40–60% of early-period supply (disclosed)
Post-emission inflation	0% — supply fixed once emission ends
Post-emission security	Transaction fees only
Token	\$CYPHER

§13 — Protocol Fee Economics

Earlier revisions of this document projected specific future \$CYPHER prices and market capitalisations. Those projections have been removed from this technical document: leading privacy and post-quantum protocol whitepapers (Zcash, Mina, Aleo) carry no price or market-cap targets in their core technical specification, and neither should this one. What follows is the fee mechanism itself — a rate structure, not a dollar forecast. Any business or valuation materials belong in a separate document, not here.

Revenue Source	Rate	Destination
MaskSwap fee	0.05% / swap	Cypher DAO treasury
ShieldLend fee	0.10% origination	Cypher DAO treasury
CypherOracle	Fixed protocol fee	Cypher DAO treasury
Bridge fee	0.05% / tx	Cypher DAO treasury
Network tx fees	Market rate	Miners (post-emission: sole security budget)

All protocol-level fees (excluding base network tx fees, which pay miners directly) flow to the Cypher DAO treasury for development, audits, and grants. This is a rate structure, not a revenue forecast, and is not financial advice.

§14 — Security Model & Attack Costs

Cypher's security is a compound of independent pillars; an adversary must compromise all simultaneously. Cost is a product, not a sum.

Pillar	Threat	Assumption	Classical Cost	Quantum Cost
CypherPOW	51% attack	Hashrate majority	>50% of GPU network	Same (PoW is PQ-safe)
CIP (STARK)	Forge existence/privacy proof	Poseidon2 collision resistance + FRI soundness	Conjectured ≥ 128 -bit	Grover-bounded ($\sqrt{\cdot}$ reduction)
Dilithium5	Forge envelope signature	MLWE + MSIS, $\dim=256$	$\sim 2^{256}$	$\sim 2^{256}$ (no known speedup)
Bridge MPC	Steal bridge funds	tECDSA 8-of-15	Control 8 validators	Control 8 validators

Replacing the prior two-row split (ring-signature forgery + Groth16 privacy break) with a single CIP row reflects that both properties are now proven by one STARK circuit sharing one soundness assumption — there is no separate, weaker privacy layer to attack.

§15 — Comparative Analysis

No comparison table survives contact with a fast-moving field for long, and this one should be read as a snapshot, not a permanent ranking. It is included because a technical reviewer should be able to see Cypher's actual position next to deployed systems, rather than take a claim of novelty on faith.

Protocol	Signatures	Privacy Proof	Trusted Setup	Default Privacy
Monero	EdDSA (not PQ)	Ring signatures + Bulletproofs	None	Yes, mandatory
Zcash (Orchard)	RedPallas (not PQ)	Halo2 (recursive)	None	Opt-in shielded pool
Mina	Not PQ	Pickles (recursive)	None (transparent)	No (state-size focus, not privacy)
Aleo	Not PQ	zkSNARK (Marlin/Varuna)	Yes — 2021 MPC ceremony	Program-defined, per transition
QRL	XMSS (hash-based, PQ)	None	None	No
Cypher (CIP, proposed)	Dilithium5 (PQ)	FRI-STARK (Poseidon2)	None (transparent)	Yes, mandatory

Based on public documentation for each project; each evolves independently of this document and should be checked against current sources before being relied on. Monero, Zcash and Aleo figures reflect their live, shipped systems; Cypher's row describes a research-stage proposal, not a deployed network — see §18.

Read plainly: as far as public documentation shows, no deployed Layer-1 currently combines post-quantum signatures with a post-quantum, mandatory-by-default privacy proof and no trusted setup. Aleo has default-adjacent privacy but keeps a trusted setup and is not post-quantum. QRL is post-quantum but has no privacy layer at all. Zcash's Orchard already proves that a transparent-setup, recursive proof system can ship in production — Cypher's claim is narrower than “first ever”: it is that CIP is the first attempt to combine that lineage of transparent-setup proving with mandatory privacy *and* post-quantum security in one L1. That is a real, specific gap being targeted, not a vague claim of superiority — and it is exactly why §4.6, §4.7, and §18 matter: the gap is only closed once CIP is actually specified, benchmarked, and audited.

§16 — Legal, IP & Regulatory Considerations

16.1 — Intellectual Property Strategy

A defensive-publication approach — publishing CIP as a pseudonymous preprint via IACR ePrint — is recommended over patent filing, to establish prior art and keep the design open while avoiding attribution to any legal identity.

16.2 — Foundation Structure

Two structures are under consideration: a Swiss Verein or Foundation, and a Cayman Foundation Company. Both require identified beneficial owners to the relevant registrar or regulator, regardless of whether the project's public branding remains pseudonymous.

16.3 — Patent Filing Constraints

For inventors based in India, a Foreign Filing License under Section 39 of the Patents Act is required before filing a patent application abroad. For non-US inventors filing with the USPTO, standard non-resident procedures and any applicable home-jurisdiction foreign filing license apply.

16.4 — Regulatory Compliance

The EU Anti-Money Laundering Regulation (AMLR), Article 79, prohibits regulated custodians from listing anonymity-enhancing tokens from July 2027 onward — a direct regulatory headwind for a privacy-by-default asset such as \$CYPHER within EU-regulated venues. It is noted here as a design constraint, not yet resolved.

§17 — Roadmap & Governance

Cypher's roadmap is presented in research-stage form — phases, not calendar dates.

- **Research & Specification** (current phase) — CIP formal definition, circuit design, whitepaper iteration.
- **Formal Verification** — academic-partner review of the CIP circuit and STARK parameters.
- **Testnet** — CypherPOW + CIP integration testing, bug bounty.
- **Mainnet Candidate** — audited release candidate, pending independent security audit.
- **Maturity** — DAO handover, fee-only security model, protocol ossification.

The core team is pseudonymous; independent security audits are planned but not yet complete, and no phase above carries a committed date. Governance votes, once Cypher DAO is live, are intended to be CIP-attested events with anonymous zk-proofed ballots, requiring 5% quorum and a 67% supermajority for constitutional changes — this mechanism itself remains a design proposal pending the CIP circuit's formal verification.

§18 — Limitations & Future Work

- CIP has no audited precedent; it is a research proposal.
- Poseidon2 is a newer hash function with a shorter cryptanalysis track record than more established hash functions.
- STARK proof sizes and verification time for the CIP circuit have not been benchmarked on real hardware.
- The emission schedule has an unresolved tail: pure annual-halving decay converges to roughly 80% of the 1,000,000,000,000 \$CYPHER target, not 100% (§12.3).
- Bridge quantum safety: tECDSA is classically secure but not post-quantum; a lattice-based threshold MPC upgrade is a future direction with no committed date.
- Open thread: an opt-in (non-mandatory) privacy mode is being considered as an alternative to the current mandatory-privacy model.
- Open thread: LP yield mechanics for wrapped bridge assets (cETH, cBTC, cSOL) are unresolved.
- Open thread: achieving Solana-comparable block speed is a goal without a concrete design yet.
- Open thread: Kaspa's GHOSTDAG is being studied as a reference for faster PoW block production.

- Formal verification of the CIP circuit is planned with academic partners but not yet scheduled.

§19 — Changelog — v1.1 → v1.4

- Replaced the separate ring-signature (existence) + Groth16 zk-SNARK (privacy) design with a single unified FRI-based STARK circuit, the Cypherpunk Inclusion Proof (CIP), removing the inconsistency between miner-side attestation and user-side privacy proofs.
- Collapsed the four-layer stack (L3/L2/L1/L0) into a three-layer stack (L2 Applications / L1 CIP / L0 Consensus).
- Removed the Groth16 post-quantum caveat and the Powers-of-Tau trusted-setup requirement — STARKs need neither.
- Added §8, Privacy Model & Key Management: mandatory default privacy, Zcash-style user-controlled viewing keys, KYC handled at fiat on/off-ramps, and explicit rejection of a Foundation-held universal decryption key as a backdoor.
- Disclosed a foundation mining allocation of 40–60% of early-period supply directly, replacing zero-premine / fair-launch framing (§12).
- Flagged an unresolved emission-schedule tail: pure geometric halving converges to ~80% of target supply, not 100% (§12.3).
- Added §16, Legal, IP & Regulatory Considerations: defensive-publication strategy, foundation-structure options, India's Foreign Filing License requirement, and EU AMLR Article 79.
- Converted the roadmap to phase-based, date-free framing; noted the team is pseudonymous with audits pending (§17).
- Expanded Limitations & Future Work with CIP-specific caveats and remaining open design threads (§18).

19.1 — v1.2 → v1.3

- Added §4.7, Concrete Parameters: a proposed field (Goldilocks), FRI blowup/query/grinding parameters, and hash choice for CIP, replacing the previously unspecified proof system with numbers that can actually be benchmarked and cryptanalysed.
- Added §4.8, Recursion & Light Clients: a recursive proof-composition strategy (in the lineage of Mina's Pickles) so per-transaction CIP proofs fold into a single block-level proof, addressing the fact that standalone STARK proofs are larger than the old ring signature.
- Added §7.1, Throughput & Comparative Benchmarks: a first illustrative TPS range and a direct comparison against Monero, Zcash Orchard, and Solana — no version of this document previously stated a throughput number at all.
- Added §10.1, Upgrade Path: zk Light-Client Bridging: proposes replacing the 8-of-15 federated MPC bridge with source-chain consensus proofs, naming the exact failure pattern (Ronin, Wormhole, Poly Network) the current design shares with past bridge losses.
- Added §11.1, Relay Decentralisation: proposes growing CypherOracle's relay set beyond 15 and adding a zk-verified DEX-read source.

- Added §15, Comparative Analysis: a sourced comparison table against Monero, Zcash Orchard, Mina, Aleo, and QRL, replacing unsupported superiority claims with a specific, checkable claim about what combination of properties is and is not currently deployed elsewhere.
- Removed all forward-looking price and market-cap projections (the former §13 table targeting \$200B by 2028) from the technical document; §13 is now Protocol Fee Economics, stated as rates rather than dollar forecasts, matching the practice of Zcash, Mina, and Aleo's own technical whitepapers.
- Removed the cover-page launch-valuation figure for the same reason.

19.2 — v1.3 → v1.4: Verification Pass

This revision stops estimating and starts checking. Every change below came from actually running the numbers or the logic, not from re-reasoning about them.

- Corrected §12's emission schedule: the table's years 3–5 never matched its own $R_y = R_0 \times k^y$ formula — they were round numbers picked to sum to 1,000B. Recomputing the formula directly gives a true total of 964,940,762,634 \$CYPHER (96.49%), superseding both the old table's false 100% and this document's own earlier, uncomputed “~80%” guess. The formula is now the stated source of truth, and the 1,000,000,000,000 figure is reframed as a ceiling, not a target.
- Re-derived §7.1's throughput claim as two separate, labelled scenarios (recursion-compressed vs. pre-recursion) instead of one blended “10–60 tx/s” range that didn't actually follow from either.
- Added §4.7.1: a real STARK benchmark (Winterfell, at CIP's exact proposed field/blowup/query/grinding parameters) on a toy circuit. Real results — 44–101 KB proofs, 4.4–7.6s single-core prove time, 11–23ms verify — confirm the “tens of KB” proof-size assumption and surface a new, more specific concern: per-transaction prover time, not block byte size, is now the more likely throughput bottleneck once recursion (§4.8) has to compose many proofs per 120-second block.
- Ran a toy, non-cryptographic protocol-logic simulation of nullifier double-spend rejection, same-block double-use rejection, and heaviest-chain fork resolution; all behaved as specified. It also showed the 6-block finality rule (§6.2) is a cost claim, not a structural impossibility — language there is now more precise about that distinction.
- None of the above is a testnet: no node implementation, network, or real Dilithium5/CIP cryptography exists yet. Both checks used stand-ins (Rescue-Prime for Poseidon2; a toy nullifier/PoW model for CypherPOW) and are labelled as such throughout.

§20 — Conclusion

Cypher's core cryptographic claim is now a single one: the Cypherpunk Inclusion Proof (CIP) proves existence and privacy together, in one FRI-based STARK circuit, wrapped by a CRYSTALS-Dilithium5 outer envelope. This removes a real inconsistency in the prior two-proof design and, because STARKs are hash-based, removes the post-quantum caveat and trusted-setup requirement that applied to the Groth16 privacy layer.

This revision moves CIP from a named idea to something with numbers attached, and then moves those numbers from estimates to checked results: a proposed field and FRI parameters (§4.7) that now have a real proof behind them (§4.7.1), a recursion strategy for keeping proofs and light clients practical (§4.8), a throughput claim re-derived instead of guessed (§7.1), an emission schedule corrected against its own formula rather than

left at a round, invented total (§12.3), and a sourced comparison against Monero, Zcash, Mina, Aleo, and QRL (§15) that states plainly what gap Cypher is actually targeting instead of asserting general superiority. Cypher remains a research-stage design with no implementation, network, or testnet yet — see §18 and §19.2 for the current, honest list of what is checked, what is stand-in, and what is still not solved.

“Ghost chains don’t Exist.”

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